A Handheld System for Three-Dimensional Mapping of Harsh Natural Caves

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***Abstract*—Exploring confined subterranean spaces, such as natural caverns, presents significant challenges due to GPS-denied conditions, severe ferromagnetic interference, and highly unstructured environment. While commercial handheld 3D scanners offer robust out-of-the-box performance, their high cost, proprietary hardware and software, and closed-source SLAM algorithms hinders their widespread use in speleological surveys. This paper presents an open-source, low-cost handheld 3D scanning device explicitly designed for harsh subterranean mapping. The system integrates a Livox Mid-360 LiDAR, an Intel RealSense D435i depth camera, and custom LED illumination, all processed on the edge via an NVIDIA Jetson Orin Nano and powered by standard 18V industrial power tool batteries. The fully 3D-printed enclosure ensures ergonomic single-handed operation and field repairability. To evaluate computational viability directly on the embedded hardware, three state-of-the-art SLAM frameworks (FAST-LIO2, FAST-LIVO, and BIEVR-LIO) were benchmarked for CPU and memory utilization. Experimental validations in real-world natural caves demonstrate that the device generates dense 3D maps with centimeter-level deviation from a commercial reference scanner. Notably, the BIEVR-LIO framework proved highly robust in geometrically sparse environments, producing point clouds highly consistent with a premium commercial scanner (median error of 4.5 cm).**

***Index Terms*—Cave Mapping, Real-world Robotics, Confined Spaces, Handled Devices, SLAM, LiDAR Odometry**

I. Introduction

THE exploration and comprehensive mapping of confined subterranean spaces, such as natural caverns, abandoned mines, and complex underground tunnel networks, present significant operational challenges and severe risks to human safety [1]. Accurate mapping of these environments is not merely an academic pursuit; it is a critical operational requirement for environmental preservation, geological surveying, and industrial compliance. Particularly in regions characterized by active mining operations, environmental licensing and risk assessment depend heavily on strict speleological classification, structural integrity monitoring, and precise volumetric analysis.

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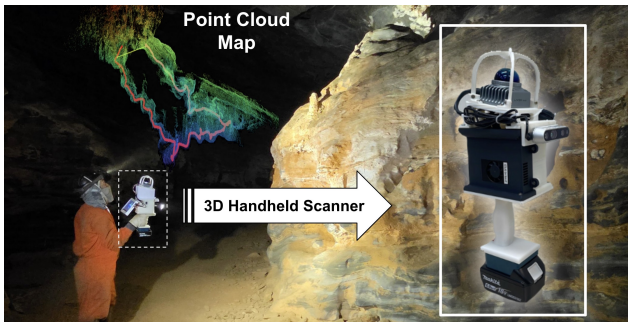


Fig. 1: Example usage of the proposed 3D mapping device in an underground cave.

In this paper, we detail the end-to-end development of this handheld scanner (Fig. 1). We describe the critical architectural, mechanical, and software decisions that enable this device to serve as a usable, robust platform for speleometry. Furthermore, we outline the practical lessons learned from conducting real-world experimental deployments in subterranean environments, providing a comprehensive guide for researchers seeking to replicate or improve upon this open-source design.

Historically, geological surveying methods in these enclosed settings has relied on traditional manually operated instruments, such as mechanical inclinometers, compasses, and handheld laser distometers. However, these traditional methods are labor-intensive, slow, and highly susceptible to cumulative errors in instruments and humans, yielding only sparse, low-resolution topological data that fail to capture the complex volumetric reality of subterranean formations. In contrast, 3D mapping using advanced Simultaneous Localization and Mapping (SLAM) techniques produces precise, high-density geometric models based on point clouds [2]. These point clouds and derived meshes are crucial for comprehensive geological studies and spatial reasoning, entirely bypassing the limitations of traditional manually operated instruments. The transition to SLAM-based surveying has revolutionized subterranean mapping, allowing for rapid, high-fidelity data acquisition. However, deploying these sophisticated technolo-

gies in extreme field conditions remains a complex engineering challenge.

Building on our prior development of multi-sensor mapping units for robotic platforms [3]–[5], this paper introduces an open-source, handheld 3D scanning device designed for human operators navigating extreme, GPS-denied subterranean terrains. To democratize high-fidelity mapping in confined spaces where robots often struggle, this 3D-printed system utilizes accessible, off-the-shelf hardware. It integrates a Livox Mid-360 LiDAR, an Intel RealSense D435i camera, custom LEDs, and an NVIDIA Jetson Orin Nano, all powered by a standard 18V power tool battery.

To rigorously validate its performance, the system was directly compared against a well-established commercial counterpart, the SLAM-100. We evaluated multiple state-of-the-art SLAM algorithms (including FAST-LIO2 [6], BIEVR-LIO [7] and FAST-LIVO [8]) directly on the edge hardware in a harsh natural cave environment (Fig. 1). These experimental sites were specifically selected for their severe ferromagnetic interference, wet surfaces, and challenging narrow passageways.

II. Related Work

The mapping of confined natural environments increasingly relies on Handheld Laser Scanners (HLS) based on SLAM, overcoming the time-consuming static setups required by traditional Terrestrial Laser Scanning (TLS) [9]. The geospatial market offers various commercial solutions, ranging from the ultra-lightweight Leica BLK2GO [10] and the RTK-integrated CHCNAV RS10 [11], to heavier hybrid systems like the FARO Orbis [12]. Other notable devices widely adopted in geological surveying include the GreenValley LiGrip [13] and the Feima SLAM-100 [14]. While these systems provide polished ergonomics, varied battery autonomy (45 minutes to 2.5 hours), and embedded computing, they present significant drawbacks for harsh environment deployments. Their closed-source, black-box algorithms prevent fine-tuning for environments with severe ferromagnetic interference, and their proprietary battery ecosystems limit field repairability and accessibility.

To address these limitations, academic initiatives have developed custom, open-source mapping units. Our previous work introduced the Autonomous Mapping and Navigation Unit (AM&NU), which evolved from robot-tethered sensor towers [3] to more compact visual-LiDAR setups [4]. However, these early iterations still relied on bulky, two-hand rig designs weighing over 3.0 kg, depended on power-hungry Intel NUCs for computation, and required external PCs via SSH for user interface (UI) interaction. While they demonstrated that low-cost setups could produce accurate 3D maps, they lacked the ergonomic refinement and standalone usability required for agile subterranean exploration.

The device proposed in this paper bridges this gap by integrating all necessary components into a single, ergonomic handheld form factor with a custom 3D-printed handle. Furthermore, it runs state-of-the-art open-source algorithms (such as FAST-LIO2) and introduces a user-friendly GUI hosted via Dev Containers, eliminating the need for complex external setups and offering a robust, fully open-source alternative to commercial black-box scanners.

III. Proposed Three Dimensional Scanner System

This section details the architecture of the proposed Spatial Pointcloud Environment Capture (SPEC) 3D scanner system. Designed primarily for GPS-denied, subterranean, and complex indoor environments, the system integrates multi-modal perception sensors with dedicated embedded hardware to execute robust online localization and dense mapping.

As illustrated in Fig. 2, the system pipeline fuses high-frequency IMU measurements, raw 3D LiDAR point clouds, and visual camera data within an embedded processor running one of the selectable SLAM frameworks such as FAST-LIO2, FAST-LIVO, or BIEVR-LIO. This integration produces online pose updates alongside a dense, global 3D point cloud representation of the mapped environment.

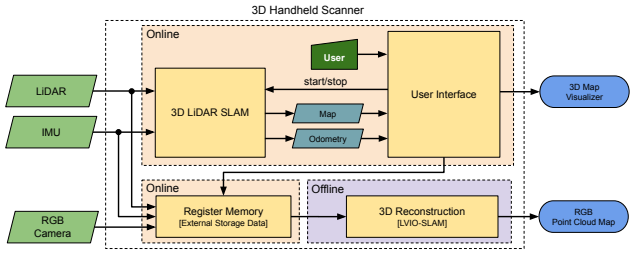


Fig. 2: Overview of the handheld 3D scanner's four main modules: 3D LiDAR SLAM, user interface, data logging, and colored map reconstruction.

A. Localization and Mapping

To achieve high-quality mapping and localization in GPS-denied and subterranean environments, the FAST-LIO2[1], FAST-LIVO[2], and BIEVR-LIO[3] frameworks were evaluated on the developed handheld scanner. The performance of these frameworks was assessed on the embedded computer (NVIDIA Jetson Orin Nano) by analyzing CPU and RAM usage, and the reconstructed point cloud. By using high-frequency inertial measurements (IMU) and fast LiDAR scans, the system is able to perform in confined environments.

1) FAST-LIO2: This is a LiDAR Inertial Odometry (LIO) framework that offers a fast, robust, and versatile approach to LiDAR navigation and mapping [6]. Its architecture employs a highly efficient, tightly coupled iterated Kalman filter that directly registers raw LiDAR points to the map. This approach allows the system to exploit subtle environmental features and adapt seamlessly to different scanning patterns found in emerging solid-state LiDARs.

The framework's second primary innovation is the integration of an incremental k-dimensional tree (ikd-Tree), which maintains the map through efficient incremental updates, dynamic rebalancing, and on-tree downsampling. This data

[1]https://github.com/hku-mars/fast_lio
[2]https://github.com/hku-mars/fast-livo2
[3]http://github.com/ethz-asl/BIEVR-LIO

structure offers superior performance over traditional options, ensuring the system remains computationally efficient even when handling large datasets. Consequently, FAST-LIO2 can achieve mapping rates up to 100 Hz on hardware-constrained platforms while providing robust and accurate pose estimation even during aggressive motion.

2) FAST-LIVO: This technique is a fast, tightly-coupled, sparse-direct LiDAR-Inertial-Visual Odometry (LIVO) system designed to achieve accurate and robust pose estimation for SLAM tasks [8]. Its framework is built upon two directly integrated subsystems: a LiDAR-Inertial Odometry (LIO) subsystem and a Visual-Inertial Odometry (VIO) subsystem. Rather than extracting computationally expensive features, the LIO subsystem registers raw LiDAR points directly to an incrementally updated point cloud map. The VIO subsystem then maximizes efficiency by reusing these mapped LiDAR points, which are appended with previously observed image patches, to align new images. This visual alignment minimizes direct photometric errors without requiring the extraction or optimization of traditional visual features, and both subsystems jointly estimate the state by fusing their respective data with IMU readings via an error-state iterated Kalman filter.

FAST-LIVO stands out by coupling sensor data at the measurement level and tracking features directly. This featureless approach drastically cuts computational overhead while delivering high accuracy in challenging, textureless, or unstructured environments. To further enhance robustness, the system incorporates a novel outlier-rejection method that dynamically filters out unstable map points. Furthermore, FAST-LIVO is highly versatile; it supports both traditional multi-line spinning LiDARs and emerging solid-state LiDARs, each with distinct scanning patterns.

3) BIEVR-LIO: To address the degradation of standard LiDAR odometry in geometrically sparse or uninformative environments, the authors incorporate a Bump-Image-Enhanced Voxel Representation (BIEVR-LIO) framework into the algorithm [7]. Environments such as long, uniform subterranean corridors often lack sufficient macroscopic geometric constraints, which can cause traditional point-to-plane odometry systems to drift or diverge. BIEVR-LIO mitigates this degeneracy by representing local surfaces as high-resolution, voxel-wise oriented height images, referred to as "bump images". Instead of approximating a surface as a simple plane, this formulation explicitly captures fine-grained, per-pixel deviations from a dominant plane, preserving subtle geometric details that would otherwise be lost.

To leverage this high-resolution map without exceeding the computational limits of the embedded hardware, the framework utilizes a map-aware, dual-resolution point sampling strategy. This metric evaluates the potential of a voxel to provide informative gradients and selectively increases the sampling point density only in regions with rich geometric detail. Furthermore, BIEVR-LIO employs a loosely-coupled state estimation approach; it computes the pose entirely from scan-to-map registration, isolating the optimization from inertial residuals to ensure that subtle geometric cues are not overwhelmed by IMU data. For speleological mapping, this capability is highly advantageous, as it allows the system to reliably utilize the natural, fine-grained roughness of cave walls to maintain robust localization where conventional geometric features fail.

IV. Equipment Design and Implementation

This section presents the physical design and system architecture of the developed handheld scanner, which weighs 2.687 kg and has an estimated total hardware cost of approximately US$2,200. It details the integration of the electronic hardware and power distribution, the ergonomic 3D-printed mechanical structure, the containerized software framework, and the target-based multi-modal sensor calibration procedure.

A. Electronic Components and Connections

The electronic components of our device and their respective power ratings are: NVIDIA Jetson Orin Nano computing module (25W), a Livox Mid-360 solid state 3D LiDAR (14W), an Intel RealSense D435i depth camera, a 7 inch HDMI touchscreen for on device visualization (5W), a custom dual output DC-DC step-down converter (95% efficiency), a mini Led headlight (18W), and a set of buttons for user interaction. Figure 3 shows the electronics diagram.

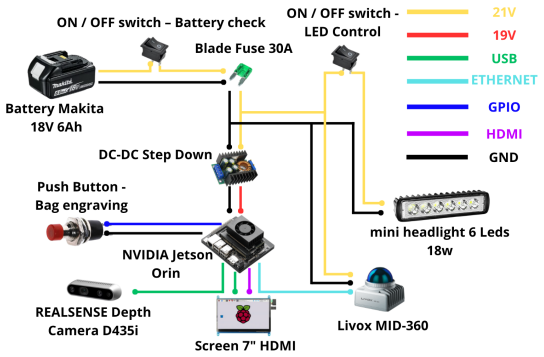


Fig. 3: Block diagram of the scanner's electronics: battery power with DC-DC conversion and Ethernet, USB, and GPIO interfaces to the embedded computer.

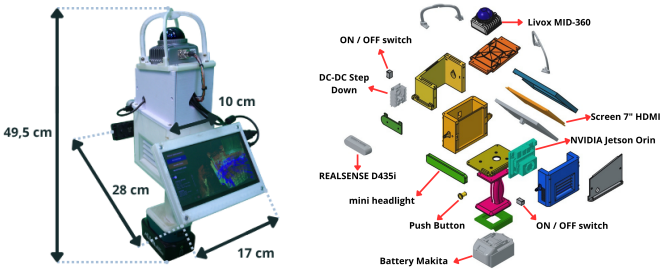
The system is powered by an 18 V, 6 Ah Makita battery pack (108 Wh), chosen for its wide availability and ease of replacement in the field. The power budget for all the components is estimated to be, in the worst case, up to 67.7W. With this Makita battery pack, the system has an estimated average autonomy of 2h10min. The battery output passes through a 30 A slowblow fuse for overcurrent protection before reaching a DC-DC step-down converter, which provides two independent power rails: 21 V and 19 V. The 21 V rail supplies the Livox Mid-360 LiDAR and the mini headlight, while the 19 V rail powers the NVIDIA Jetson Orin.

Two SPST switches and one push button are integrated into the device for direct user interaction. The first one is the device main switch, connecting or disconnecting the battery from the power circuit. the second controls the mini headlight, which is composed of 6 LEDs, used to illuminate the environment during low and light operation inside caves and confined

structures; a thin plastic layer was attached over the headlight to act as a light diffuser, preventing the images captured by the camera from becoming overexposed. The third is a momentary push button wired to a GPIO pin of the Jetson Orin, allowing the operator to start the mapping process and begin or stop recording bags directly, as an alternative to the screen's touch interface. The Livox Mid-360 communicates with the Jetson Orin via Ethernet, while the intel RealSense D435i connects through USB. The 7-inch display is powered directly by the Jetson Orin's USB port, enabling on site visualization of the point cloud and system status without requiring an external computer or SSH connection.

B. Mechanical Design

All structural components were manufactured using PLA (polylactic acid) filament through 3D printing. The Printed structure is composed of ten components: base, Jetson enclosure, circuit enclosure, LiDAR mount, LiDAR cover, Screen front and back panels, camera support and handheld mount, totaling approximately 831,87g and requiring around 20 hours of print time in a Creality K1C printer. The final assembled device weighs 2,687 kg. The dimensions of the device and the exploded 3D view if the components are depicted in Fig. 4.



(a) (b)

Fig. 4: (a) External dimensions of the assembled 3D scanner. (b) Exploded view of the component arrangement.

The enclosure consists of two main components. One holds the NVIDIA Jetson Orin, while the other contains the DC-DC step-down converter, cables, and circuit connections. A pistol grip handle is attached under the main housing, enabling one-handed operation, with a Makita battery attached to its base. The Livox Mid-360 is mounted on the top of the structure, tilted 10 degrees forward to optimize point capture in front of the operator. The Intel RealSense D435i is installed on a dedicated mount directly below it. An articulated arm supports the 7 inch HDMI display, allowing the operator to adjust the viewing angle during use. The mini headlight is mounted on the front of the structure, adjacent to the depth camera. The custom printed housing for the Jetson features small ventilation openings for heat dissipation; these are angled to hinder the entry of liquids ingress and provide additional protection for the equipment during operation in cave environments.

C. Software Architecture

Figure 5 details the software stack, organized in sucha a way that the application is decoupled from the embedded platform and the device boots straight into an operational state.

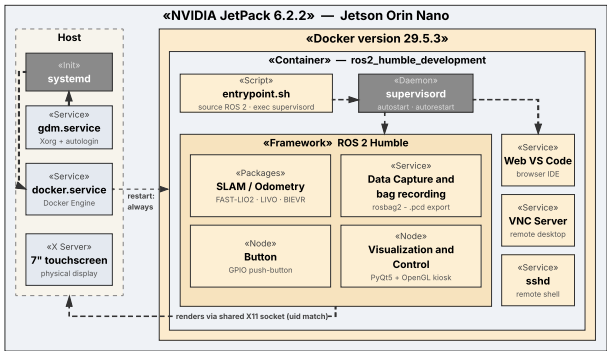


Fig. 5: Software stack: the host OS handles display and input; a ROS 2 Humble container runs acquisition, odometry, and logging.

The host runs NVIDIA JetPack and its init system starts the display manager, which logs in automatically and brings up the X server driving the touchscreen, and the container engine, which starts the container at every boot through a restart policy. Powering the device on is the only action required to begin scanning. Inside the container, a process supervisor launches every component with automatic restart, so a component that dies recovers without intervention. The ROS 2 Humble workspace is bind-mounted from the host, so code, recorded bags, and exported maps survive container rebuilds.

The odometry back-ends are interchangeable packages, allowing FAST-LIO2, FAST-LIVO, and BIEVR-LIO to be swapped through a single launch target and compared in Section V on identical hardware and recorded streams. Data capture stores the raw sensor streams as MCAP bags and exports the dense map as a point cloud, so a single descent can be reprocessed offline with different estimators. The button node publishes a GPIO push-button as a topic, so mapping and recording can be toggled without the touchscreen. Visualization and control renders the map on the touchscreen, giving the operator immediate feedback on coverage underground. A web IDE, a remote desktop, and an SSH endpoint expose the running system over Wi-Fi for field debugging.

D. LiDAR/Camera Calibration

To enable accurate sensor fusion between the LiDAR and the camera, a rigorous calibration procedure was implemented. First, the camera's intrinsic parameters (f_x, f_y, c_x, c_y) and radial/tangential distortion coefficients (k_1, k_2, p_1, p_2) were estimated using an asymmetric circle grid pattern. With the intrinsics established, the extrinsic transformation matrix between the two sensors was calculated using the FAST-Calib package [15].

The calibration was performed using a custom-scaled planar target (1.3 m $\times$ 0.9 m) featuring circular cutouts and fiducial markers, as depicted in Fig. 6. A total of five distinct captures, consisting of recorded bags and corresponding static images, were acquired from various perspectives to provide robust spatial constraints. For each dataset, the raw point cloud was pre-processed using the provided script

`distance_filter_tool` utility to precisely isolate the calibration target by defining its depth boundaries.



Fig. 6: FAST-Calib planar target (1.3 m $\times$ 0.9 m) with circular cutouts and fiducial markers for LiDAR-camera calibration.

Once the target was extracted across all five captures, a multi-scene optimization routine was executed. This routine successfully converged, yielding a final translation vector from the LiDAR to the camera of $\mathbf{t}_{cl} = [0.163, 0.132, 0.017]^T$ meters, which aligns with the physical mechanical assembly of the scanner. The accuracy of this extrinsic transformation was validated by a final optimization Root Mean Square Error (RMSE) of 0.061 meters, ensuring precise spatial alignment for the subsequent projection of RGB data onto the structural point cloud.

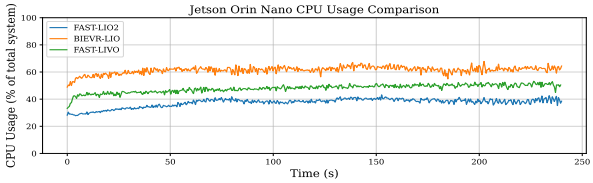
V. Experiments and Results

The device was evaluated in a real natural cave of significant size, located in the region of Sete Lagoas, in the state of Minas Gerais, Brazil. The Lapa do Boi cave have an extension of $1{,}410$ meters, with different height profiles and the presence of water, dust, and other natural structures. We also used the commercial SLAM-100[4] LiDAR scanner as a reference to compare the estimated point clouds.

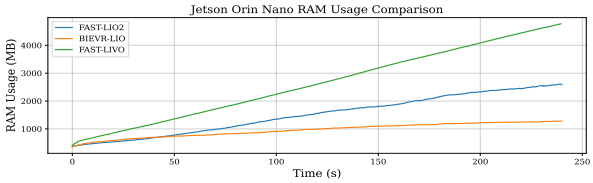
To ensure statistical reliability, computational performance was evaluated during live real-time operation by averaging data point-by-point across five independent 240-second trials for each algorithm. Performance was assessed online, on the handheld's embedded NVIDIA Jetson Orin Nano (6 cores, 8 GB RAM) configured in the MAXN SUPER power mode. CPU and RAM usage were collected using a custom Python script built on top of the *psutil* library, which sampled each package's processes at 2 Hz throughout every trial. CPU usage was normalized by the number of available cores to express it as a percentage of total system capacity, while RAM usage was recorded as the resident set size (RSS) of the corresponding processes, in megabytes. All experiments were conducted on a system running Ubuntu 22.04.5 LTS with ROS2 Humble.

During online edge processing, the CPU metrics (Fig. 7a) indicate that FAST-LIO2 is the most efficient, maintaining an average usage of 38%. FAST-LIVO averaged 48% (52% peak), while BIEVR-LIO imposed the highest computational load, averaging 62% (68% peak). However, the critical bottleneck for live deployment emerged in memory management. As shown in Fig. 7b, FAST-LIVO exhibited aggressive RAM accumulation, surging from 400 MB to nearly 4800 MB. This

[4]https://feimarobotics.net/products/3d/slam100



(a) CPU usage.



(b) Memory consumption.

Fig. 7: CPU and Memory usage of the evaluated SLAM algorithms on the NVIDIA Jetson Orin Nano.

implies in frequent memory access which introduces severe system latency, impacting real-time processing capabilities and degrading visualization fluidity, rendering it unviable for live feedback. In contrast, BIEVR-LIO demonstrated highly stable memory management on the Jetson (peaking at 1300 MB), while FAST-LIO2 showed moderate accumulation (reaching 2600 MB).

To estimate the device's accuracy, the point clouds reconstructed using FAST-LIO2 and BIEVR-LIO were compared to the point cloud acquired with the commercial SLAM-100 scanner.

The point clouds were first aligned by manually selecting four pairs of corresponding points between each reconstructed point clouds and the reference data. Fine registration was then performed using the Iterative Closest Point (ICP) algorithm [16] implemented in CloudCompare. The ICP registration used an RMS difference threshold of 1.0×10^{-5} as the convergence criterion and an expected final overlap of 100%. Registration was performed using the complete point clouds, without restricting the process to selected overlapping regions.

Then, the Cloud-to-Cloud (C2C) distance [17] was computed to quantify the geometric discrepancies between each reconstructed point cloud and the reference data. All processing steps were performed using CloudCompare.

The point-cloud comparison between the onboard algorithm that achieved the lowest median C2C distance relative to the reference (BIEVR-LIO, 4.5 cm against 6.9 cm for FAST-LIO2) and the commercial solution (SLAM100) can be observed at Fig 8. The error histogram of this comparison is also shown in Fig 9. The histogram is clustered near zero, with a median distance of 4.5 cm and a mean error of 10 cm. Furthermore, approximately 87% of the points exhibit a geometric discrepancy below 18.2 cm, while all points under 2 standard deviations

TABLE I: Cloud-to-Cloud (C2C) distance error metrics.

Comparison Pair	**Pts.** ($\times 10^6$)	**Median** (m)	1σ (m)	2σ (m)	3σ (m)
FAST-LIO2 / BIEVR-LIO	51.600	0.034	0.060	0.193	0.339
BIEVR-LIO / SLAM100	352.750	0.045	0.082	0.345	3.183
FAST-LIO2 / SLAM100	322.208	0.069	0.120	0.333	3.678

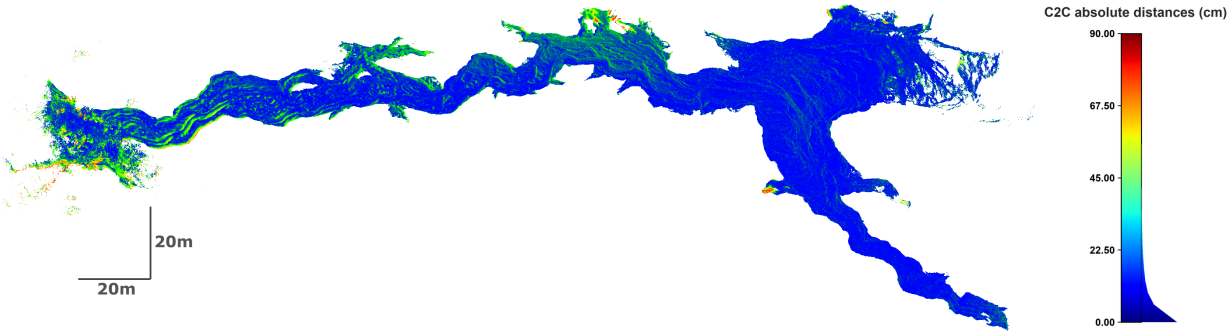


Fig. 8: Point-cloud comparison between the BIEVR-LIO map vs the commercial solution SLAM100 map of the Lapa do Boi cave (Extension of 1.4 km, with an area of 39 km^2). The color bar depicts the absolute distances in centimeters.

(95%) remain below 34.47 cm. Only a small fraction of the points fall into the long error tail, which is primarily associated with regions affected by occlusions, vegetation and differences in the trajectories followed during acquisition.

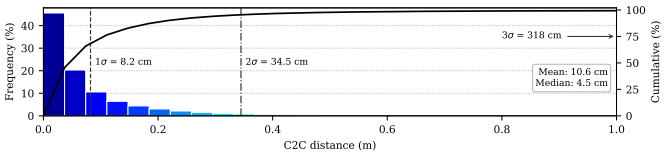


Fig. 9: C2C distance distribution between the BIEVR-LIO and SLAM-100 point clouds

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VI. Conclusion

This paper presented a novel, open-source handheld 3D scanning system engineered specifically for the severe constraints of natural subterranean environments. By integrating cost-effective, off-the-shelf hardware—including a Livox Mid-360 LiDAR and an NVIDIA Jetson Orin Nano—with standard 18V power tool batteries, the proposed device democratizes high-fidelity speleological mapping. Real-world experimental validation in harsh, GPS-denied cave systems demonstrated that the device can successfully execute state-of-the-art SLAM algorithms directly on the edge. Notably, the BIEVR-LIO framework delivered robust, stable performance in geometrically challenging corridors, yielding point clouds highly consistent with premium commercial alternatives like the SLAM-100.

While the current prototype effectively overcomes the closed-source and high-cost limitations of existing commercial scanners, field testing identified areas for mechanical and operational improvement. Future iterations will focus on reducing overall device weight, optimizing thermal dissipation for the sealed enclosure, and implementing adaptive illumination strategies to prevent camera exposure issues. Furthermore, integrating dynamic SLAM parameter configuration will help prevent tracking loss in the most narrow and uninformative subterranean passages, further advancing this accessible paradigm for extreme environment mapping.

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